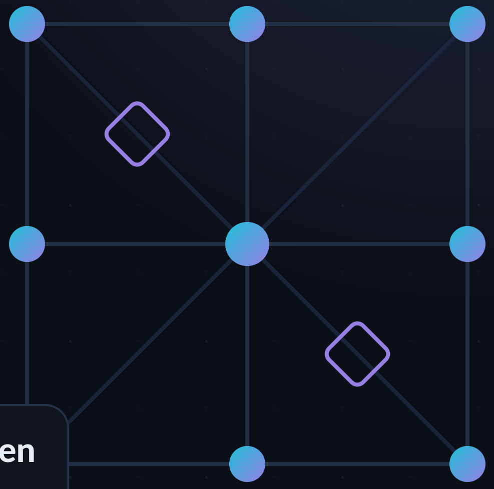
SEA QUANTATHON 2026 · DEEP-TECH QUANTUM COMPUTING

SYNDRA



The error-correction layer that makes quantum computers reliable enough to solve real problems

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SDG 3 · 7 · 9 · 13

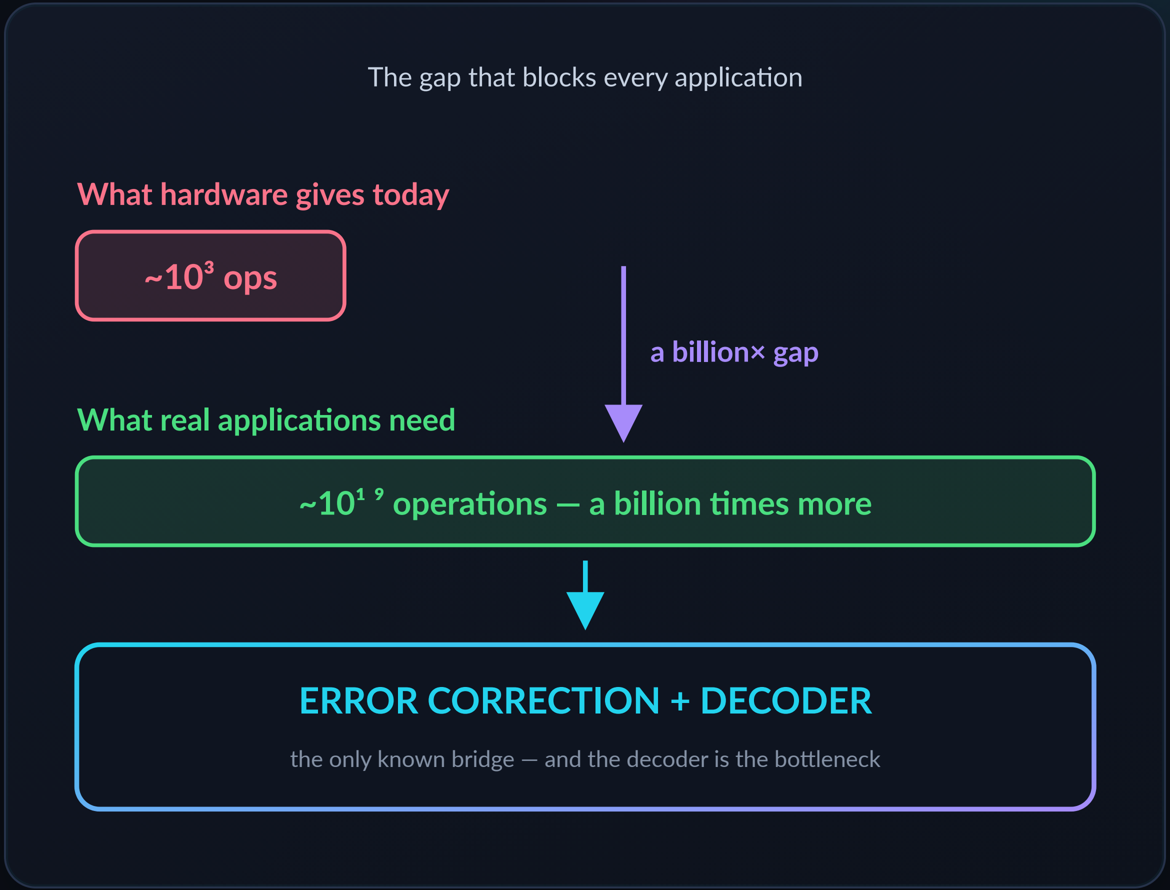
$16.3B market by 2030

Open-source · hardware-agnostic

THE PROBLEM — AND WHY NOW

Quantum computers are too error-prone to be useful

- Today's quantum chips make **an error roughly every 1,000 operations**. Real applications need **billions** of operations in a row.
- The fix is **error correction**: measure error signals every cycle, and a **decoder** works out what went wrong — in **microseconds**, or the computation is lost.
- **Urgency:** hardware is scaling **now** — $13B of public money committed, 100+ logical qubits announced in 2026. The decoder is what's holding the field back.
- **Without a working decoder, none of quantum's promised impact** — medicine, clean energy, climate — ever arrives.



No decoder, no bridge. **This is the blocker we work on.**

WHAT WE BUILD

A smarter proofreader for quantum computers

A quantum computer is like a page of text where letters randomly flip while you read. **Error correction is the proofreader** that spots and fixes them before you see the page. **We build that proofreader** — and we make it smarter using quantum machine learning.

THE PRODUCT

A software decoder

Not hardware. A **software library (SDK)** that plugs into a quantum computer's control system — the way a graphics driver plugs into a PC.

WHAT IT DOES

Reads alarms, fixes errors

Every cycle the machine emits error signals. Our decoder reads them and outputs **what to correct** — continuously, while the program runs.

WHY OURS IS DIFFERENT

It learns the machine

Every quantum chip has its own noise fingerprint. Ours is **trained per machine**, so it adapts instead of assuming a textbook error model.

Where we sit: we don't build the machine — we make it trustworthy

Quantum chip
any maker, any technology

→ error alarms →

SYNDRA decoder
software · learns this machine

→ corrections →

Reliable computation
runs long enough to matter

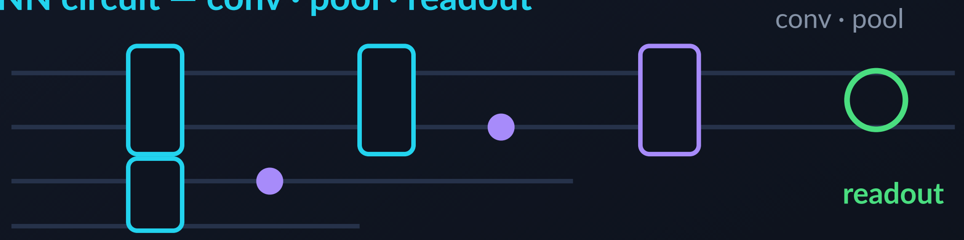
→

Real applications
medicine · energy · climate

Quantum machine learning, used where it actually fits

- **Real QEC stack, not a toy:** surface code simulated in **Stim**, decoders in **PennyLane + PyTorch**, matched against **PyMatching** (MWPM) — the same tools Google and Riverlane use.
- **QCNN, chosen for a reason:** Cong et al. proved it is **barren-plateau-free** — gradients survive as qubits grow, where generic quantum ML fails to train at all.
- **O(log N) parameters:** ~4,000 vs millions in classical neural decoders — cheaper to train and re-tune per machine.
- **Patch design keeps it scalable:** qubits per circuit stay **constant** as the code grows — only the number of circuits scales.

QCNN circuit — conv · pool · readout

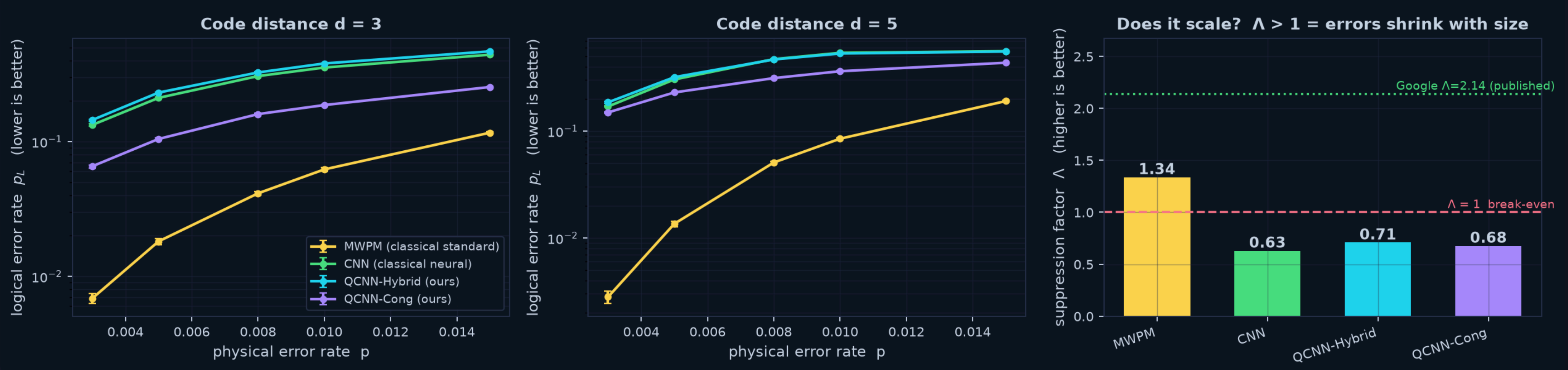


No quantum-washing: the quantum layer does the pattern-matching; classical layers handle what classical does better. We report where quantum **does not yet win** — see next slide.

BENCHMARK

Among learned decoders, quantum wins

Measured on identical fresh test data — 20000 shots per point



Same data for all four

Identical fresh test set per point; test seed disjoint from training — **no leakage**.

Paper-exact metrics

Logical error rate, Λ suppression, error-per-cycle — from **Google, Nature 2024**.

Quantum beats classical ML

At **equal training budget**, our quantum decoder **Λ=0.71** vs classical neural **0.63**.

Honest finding

MWPM — a 40-year-old specialist algorithm — still leads overall. **We publish that.**

All points measured on fresh test data, 20k shots each. QCNN-Cong measured on an earlier architecture revision (2026-07-25); all others on the current build.

SOCIAL IMPACT & UN SDGS

Error correction is what unlocks quantum's social value

SDG 3

Good Health & Well-being

Molecular simulation for drug discovery — industry projections cite up to **60% shorter** development cycles.

SDG 7

Affordable Clean Energy

Better batteries, solar cells and catalysts — projected **30–40%** efficiency gains in renewables research.

SDG 9

Industry & Innovation

Directly ours. We build core infrastructure for a resilient, next-generation computing industry.

SDG 13

Climate Action

Higher-fidelity climate and materials modelling — projected **40%** accuracy improvement in climate simulation.

THE HONEST LINK None of those benefits arrive on today's error-prone machines. **Error correction is the enabling layer** — and our open-source approach also serves **SDG 10**: keeping quantum access open to emerging hubs like Southeast Asia, not locked behind proprietary stacks.

Error correction works — what we build → Quantum computers usable — long enough to solve real problems → **SDG 3 · 7 · 13 outcomes become reachable** — medicine · clean energy · climate

SDG impact figures are industry projections (UNDP Deep Tech Series; TCS; Pasqal; WEF 2024), not our measurements.

MARKET & BUSINESS MODEL

Who pays, how we ship, how we earn

MARKET BY 2030

$16.3B

From $2.2B in 2025, 33.7% CAGR. Every machine in it needs a decoder.

QEC FUNDING, 18 MONTHS

$400M

Raised by QEC-focused vendors alone — Riverlane, Q-CTRL, Quantum Machines.

PUBLIC COMMITMENT

$13B

US $2B, China $10B, EU €1B — QEC sits on every national roadmap.

CUSTOMERS

- **First customer: our challenge sponsor** — a photonic hardware maker whose stack we already run on.
- Then: other hardware makers, cloud QC platforms, national labs.

HOW WE SHIP

- Decoder SDK integrated into the partner's control stack.
- Open-source core for adoption; paid integration & tuning on top.

REVENUE

- Licensing by decode volume in production.
- Annual support + per-hardware tuning contracts.
- Co-development & national grant funding.

Sources: market.us, Precedence Research (market); TheQuantumInsider (QEC funding); NIST / Dept. of Commerce, EU Quantum Flagship (public funding).

COMPETITIVE LANDSCAPE

Where we sit — and the lane we take

DECODER	LATENCY	ADAPTS TO A MACHINE'S NOISE	PARAMETERS	OPENNESS
MWPM (classical standard)	medium	no	n/a (algorithmic)	mature tooling
Google AlphaQubit	~63µs	yes	millions	proprietary
Riverlane Deltaflow	**6.5µs**	yes	large	proprietary, $75M+ raised
SYNDRA (ours)	target: real-time	**yes, per machine**	**O(log N)**, ~4k	**open-source core**

OUR WEDGE The fast decoders are **closed and hardware-locked**. Smaller hardware makers — including in Southeast Asia — cannot buy in. We are the **open, adaptable, low-parameter** option for everyone else.

O(log N)
PARAMS VS MILLIONS

0
HARDWARE LOCK-IN

100%
OPEN-SOURCE CORE

A concrete path from demo to deployed product

1 2 3 4

Months 1–3

Integrate

- Plug our SDK into the sponsor's control stack.
- Import their real device noise model.
- Reproduce their current decoder as the baseline to beat.

Months 4–6

Train & beat baseline

- Train on their noise fingerprint, not a textbook model.
- Target: cross $\Lambda > 1$, then beat MWPM.
- Deliver a reproducible benchmark report.

Months 7–9

Validate on hardware

- Run on the sponsor's real QPU via their cloud.
- Measure end-to-end decode latency in situ.
- Joint technical publication.

Months 10–12

Ship

- Production-grade SDK release, open-source core.
- Convert sponsor into **first paying customer**.
- Scale to further distances & second hardware partner.

DE-RISKED Months 1–3 are **already built**: working pipeline, 4 decoders, paper-exact metrics, GPU training, running today on the sponsor's cloud.

TEAM & THE ASK

Why this team, and what we need

Truong Le Gia Khanh
LEAD · QUANTUM RESEARCH

PhD candidate, University of Sydney. QEC theory, architecture design, scientific direction.

Chu Tuan Kiet
QUANTUM ALGORITHMS

University of Technology Sydney. QCNN circuit design, surface-code modelling.

Hoang Dinh Duy Anh
SOFTWARE ENGINEERING

FPT University. Pipeline, SDK architecture, reproducible test infrastructure.

Nguyen Gia Hien
ML & BENCHMARKING

FPT University. Model training, GPU optimisation, benchmark methodology.

4
DECODERS BENCHMARKED

50+
AUTOMATED TESTS, ALL PASSING

100%
OPEN-SOURCE & REPRODUCIBLE

12 mo
TO FIRST PAYING CUSTOMER

THE ASK Incubation support, compute credits, and seed funding — to turn a working benchmark into the **open error-correction layer** the quantum industry runs on. **Let's talk.**

Quantum computing will only matter if it becomes **reliable**.

We build the layer that makes it reliable.